

QUANTITATIVE EQUITY RESEARCH

# Can Fundamental Acceleration Predict Mid-Cap Stock Outperformance?

A Point-in-Time Factor Model  
Using Bloomberg Terminal & Python

U.S. mid-cap equities | Historical cohorts of 2014, 2019, and 2022  
36-month cumulative total return  
Growth 50% · Quality 30% · Valuation 20%

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# 1 Executive Summary

This study tests whether a point-in-time composite factor model can identify U.S. mid-cap companies that subsequently outperform comparable equities over a 36-month holding period. Bloomberg Terminal supplies the historical financial inputs and return data; Python implements universe construction, factor scoring, portfolio formation, and performance evaluation. The evidence varies materially by cohort: 2019 provides the strongest result, 2022 delivers relative outperformance without an unusual random-portfolio outcome, and 2014 is a meaningful failure case.

The model assigns 50% to Growth, 30% to Quality, and 20% to Valuation. Its six inputs combine revenue growth acceleration and forward earnings revisions with ROIC, free cash flow margin, and two growth-adjusted valuation measures. Factors are winsorized, percentile-ranked, and standardized within historical GICS sectors before they enter the composite score. The study evaluates this combined model; it does not isolate the predictive contribution of revenue acceleration.

The three portfolios are formed at the end of 2014, 2019, and 2022 and evaluated during 2015–2017, 2020–2022, and 2023–2025, respectively. Each Top 25 portfolio begins equally weighted and is held for 36 months without monthly rebalancing. Comparisons include the Bottom 25 factor-ranked companies, the S&P MidCap 400, and 100 separate random 25-stock portfolios drawn from the same eligible scoring universe in each cohort.

The Top 25 earned 36-month cumulative total returns of 13.43%, 58.40%, and 82.80% in the 2014, 2019, and 2022 cohorts. Corresponding Random Portfolio Means are 25.77%, 7.74%, and 65.28%. Only the 2019 selection exceeded all 100 sampled random portfolios, with an empirical p-value of 0.0099. The 2022 selection beat 73% of the random portfolios; the 2014 selection beat only 16%. Risk statistics and the small number of historical observations limit any broader conclusion.

## 2 Research Question & Motivation

The central question is whether accelerating fundamentals, business quality, and growth-adjusted valuation can jointly help distinguish future outperformers within U.S. mid-cap equities. This is economically relevant because stock selection concerns the return on invested capital relative to available alternatives. A company need not cross a large-cap classification threshold to generate an attractive investment outcome; equally, growth in corporate size is not sufficient evidence of superior stock returns.

Revenue acceleration measures whether year-over-year growth is improving relative to the prior quarter, rather than simply whether revenue is growing. Such change may contain information about the direction of a business. The model pairs it with changes in forward earnings expectations, measures of profitability and cash generation, and valuation relative to growth. Together, these factors test whether improving fundamentals are more useful when supported by strong profitability and reasonable valuation. The historical test assesses their combined ranking, without establishing a causal mechanism or a universally reliable investment edge. Prior research has documented return relationships associated with momentum, profitability, valuation, and broader multifactor characteristics, providing context for the signals combined in this model (Fama & French, 2015; Novy-Marx, 2013; Tomtosov, 2024; van Vliet et al., 2025).

### 3 Data & Historical Universe

Bloomberg Terminal is the primary financial-data source. Historical screening and Excel/BQL-style workflows support point-in-time universe reconstruction, financial characteristics, and subsequent monthly total returns. The research universe consists of U.S.-domiciled common stocks. Historical cohort formation is intended to align security eligibility and company information with each screen date, rather than applying a present-day universe to earlier periods. Historical screens included both active and subsequently delisted securities to reduce survivorship bias. Factor inputs were restricted to information available at each cohort's formation date, while subsequent returns were used only for evaluation.

#### Historical cohorts and eligible companies

| Cohort | Screen date | 36-month evaluation window | Scoring universe |
|--------|-------------|----------------------------|------------------|
| 2014   | 31 Dec 2014 | Jan 2015–Dec 2017          | 823              |
| 2019   | 31 Dec 2019 | Jan 2020–Dec 2022          | 810              |
| 2022   | 30 Dec 2022 | Jan 2023–Dec 2025          | 840              |

Source: Author calculations using Bloomberg Terminal data. Scoring-universe counts reflect the factor-availability rules after the market-cap and liquidity screens.

Market capitalization is converted to U.S. dollars using screen-date foreign-exchange rates before economic companies are ranked. The mid-cap band is defined as company market-cap ranks 501–1500. A 3-month average daily dollar volume above \$5 million is then required. Companies are not reranked after that liquidity filter. The resulting eligible universes contain 944, 953, and 962 companies, respectively, before the factor-availability requirements reduce them to the scoring universes shown above.

#### Economic-company deduplication

Multiple securities can represent a single economic company. The notebook groups records with identical U.S.-dollar market capitalization and selects a canonical record using factor coverage, non-legacy ticker status, and raw rank. Confirmed alias groups address additional duplicates, and available values in those records fill missing fields. Duplicate-size and near-duplicate audits support these identity decisions. The company rank band is applied after this reconstruction, so the portfolio selection operates at the economic-company level.

Ticker aliases are retained for return matching and audit checks. Each company's return series uses the canonical ticker selected during deduplication; aliases are not chosen on the basis of later investment performance. The saved return-panel checks report 36 months for every scored company and no missing monthly returns in the final panels. That coverage supports the calculation, while the historical identity and return-matching assumptions remain relevant limitations.

#### Historical classification and data access

Historical Global Industry Classification Standard (GICS) sectors define the peer groups used for factor normalization. The reconstruction includes the sector structure applicable in 2014 and reversions for classifications affected by the 2023 changes. Missing classifications and sector counts are checked before scoring. This historical mapping matters because a company's appropriate peer group can change over time.

Licensed Bloomberg raw data and security-level source workbooks are not reproduced publicly. The report presents aggregate methodology, portfolio outcomes, and the notebook's final charts. The research record remains the finalized notebook, with private financial-data inputs underlying the historical screens and return calculations.

## 4 Factor Model

The composite uses the same 50% Growth, 30% Quality, and 20% Valuation specification across the three cohorts. These are the notebook's fixed model weights, not a set of weights fitted to the returns reported here. Within each pillar, two available factor scores receive equal weight; when only one is available, it carries that pillar's full weight.

| Pillar    | Weight | Factors                                              | Preferred direction |
|-----------|--------|------------------------------------------------------|---------------------|
| Growth    | 50%    | Revenue Growth Acceleration<br>3M BF12M EPS Revision | Higher              |
| Quality   | 30%    | ROIC<br>FCF Margin                                   | Higher              |
| Valuation | 20%    | BF12M PEG<br>Growth-Adjusted EV / Forward Sales      | Lower               |

### Factor definitions and availability

Revenue Growth Acceleration is current year-over-year quarterly revenue growth minus the preceding quarter's year-over-year revenue growth. The two growth rates compare the latest quarter with four quarters earlier and the prior quarter with five quarters earlier. Nonmissing, nonzero revenue denominators are required. The 3M BF12M EPS Revision is current Bloomberg forward 12-month earnings per share divided by its estimate three months earlier, minus one; its earlier estimate must be nonmissing and nonzero.

Quality uses Bloomberg ROIC (return on invested capital) and FCF Margin (free cash flow margin) directly. Valuation uses BF12M PEG and Growth-Adjusted EV / Forward Sales. The latter divides enterprise value by next-year estimated sales, then divides that ratio by forecast sales growth from next year to year two. Next-year estimated sales must be positive. Nonpositive forecast sales growth makes the growth-adjusted EV/sales metric missing, and nonpositive PEG is also treated as missing.

A company must have at least one available factor in every pillar to enter the scoring universe. Missing observations are not assigned an artificial favorable value. Available component scores are averaged within each pillar; a company missing both inputs in any pillar is excluded. The pillar weights therefore remain 50/30/20 even when the number of available underlying factors differs across companies.

### Transformation and composite construction

Within each cohort, each raw factor is winsorized at the 1st and 99th percentiles of its nonmissing observations. Values beyond those thresholds are capped. The capped values are then percentile-ranked across the full scoring universe. This sequence reduces sensitivity to extreme magnitudes while preserving a cross-sectional ordering.

Those full-universe percentile ranks are standardized within historical GICS sectors: the sector mean is subtracted and the result divided by the sector's population standard deviation (ddof = 0). A valid observation in a group with only one observation or no dispersion receives a neutral z-score of zero; missing values remain missing. Valuation z-scores are reversed so that lower valuation measures score better.

The final composite equals  $0.50 \times \text{Growth} + 0.30 \times \text{Quality} + 0.20 \times \text{Valuation}$ . Higher composite scores rank first. Tied composite ranks are resolved by existing row order using the notebook's "first" rule, with selection-boundary checks for the Top 25 and Bottom 25. Sector normalization applies to factor scores; it does not impose sector portfolio weights.

## 5 Portfolio Construction & Evaluation

### Selection and holding-period mechanics

For each cohort, the 25 highest-ranked companies form the Top 25 portfolio and the 25 lowest-ranked companies form the Bottom 25 comparison portfolio. Each position starts at an equal 4% weight. Both portfolios are held for 36 months without monthly rebalancing. As constituent values change, portfolio weights drift with their realized performance.

Monthly total returns are compounded separately for each company. The average of those constituent wealth paths represents an equally weighted initial investment held through time. Final portfolio wealth minus one is the 36-month cumulative total return. The method therefore evaluates the return on the initial allocation, rather than repeatedly resetting every constituent to the same monthly weight.

Cumulative total return is the primary outcome throughout this report. It measures the total gain or loss over the complete three-year holding period and is not an annual return. The notebook also calculates compound annual growth rate (CAGR) as  $(1 + \text{cumulative return})$  raised to the one-third power, minus one. CAGR is a separate supplementary annualized measure; it is not the return displayed in the main results table or either figure.

### Random portfolios from the same scoring universe

For each cohort, 100 separate random portfolios of 25 distinct stocks are drawn from the same eligible scoring universe used by the factor model. Selection is without replacement within each portfolio, so a stock cannot occupy two positions in one draw. A company may appear in more than one of the 100 separate portfolios. Each random portfolio is equally weighted at formation and held for the same 36 months without monthly rebalancing.

A 36-month cumulative total return is calculated for each random portfolio. The reported Random Portfolio Mean is the arithmetic mean of those 100 portfolio-level 36-month cumulative total returns. It is not an annualized return, a single randomly selected portfolio, or the return of one portfolio containing all sampled stocks. The notebook uses a reproducible random seed of 42 plus the cohort year.

This benchmark places the model-selected 25 stocks against equally sized selections from the same opportunity set under the same holding rule. The sampled return distribution also supports the empirical test in Section 7. A return above the Random Portfolio Mean and an unusually high position within that distribution are distinct findings; the 2022 cohort illustrates why both matter.

### Index and low-score comparisons

The S&P MidCap 400, identified as MID Index in Bloomberg, supplies an external mid-cap benchmark. Its monthly return sheets are checked for 36 observations and compounded over the same cohort windows using the shared wealth-path function. This comparison provides index context alongside the more directly matched random portfolios drawn from the model's own scoring universe.

The Bottom 25 comparison asks whether the high-score selection outperforms the low-score selection under identical initial weighting and holding-period rules. It is a comparison between two long-only portfolios, not a reported long-short trading strategy. Together, the random portfolios, index, and Bottom 25 address different aspects of relative investment performance.

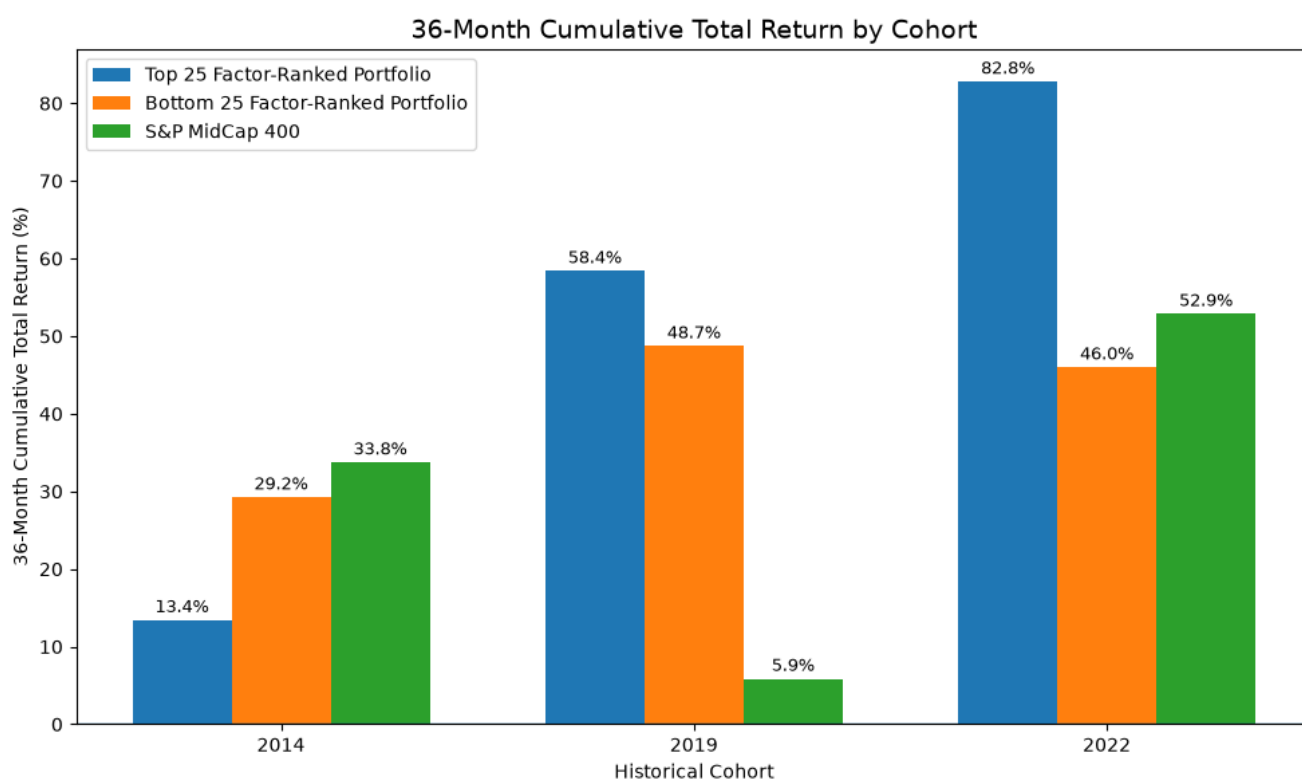
## 6 Results

Table 1 reports 36-month cumulative total returns. Cohort labels refer to portfolio formation years; the return windows begin the following January. The outcomes should be read cohort by cohort because the differences across periods are central to the evidence.

**Table 1. Portfolio and benchmark returns over 36 months**

| Cohort | Top 25 | Random Portfolio Mean | S&P MidCap 400 | Bottom 25 |
|--------|--------|-----------------------|----------------|-----------|
| 2014   | 13.43% | 25.77%                | 33.75%         | 29.25%    |
| 2019   | 58.40% | 7.74%                 | 5.87%          | 48.72%    |
| 2022   | 82.80% | 65.28%                | 52.87%         | 46.00%    |

Random Portfolio Mean = arithmetic mean of 100 separate 25-stock portfolios' 36-month cumulative total returns. Source: Author calculations using Bloomberg Terminal data. All returns in Table 1 are cumulative, not annualized.



**Figure 1. Top 25, Bottom 25, and S&P MidCap 400: 36-month cumulative total returns by cohort.** Source: Author calculations using Bloomberg Terminal data; display labels are rounded to one decimal place.

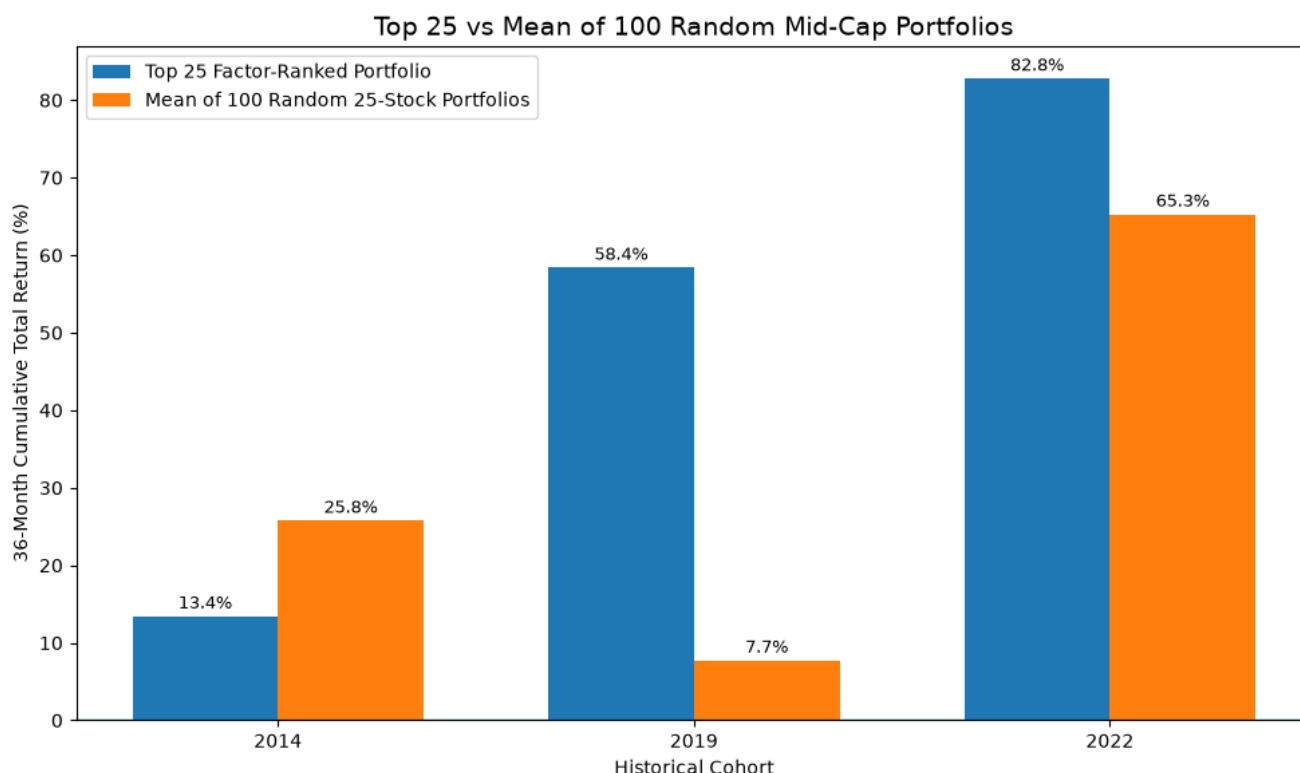
### 2014 cohort | January 2015–December 2017

The Top 25 returned 13.43%, below the Random Portfolio Mean of 25.77%, the S&P MidCap 400's 33.75%, and the Bottom 25's 29.25%. The analysis shows a Top 25 shortfall of 20.32 percentage points versus the index and 15.82 percentage points versus the Bottom 25. Positive absolute returns therefore did not translate into successful relative stock selection. This cohort is a substantive failure of the model in the tested period and limits any claim of consistent selection ability.

## 2019 cohort | January 2020–December 2022

The 2019 selection produced a 58.40% 36-month cumulative total return, compared with 7.74% for the Random Portfolio Mean, 5.87% for the S&P MidCap 400, and 48.72% for the Bottom 25. The recorded return spreads were 52.54 percentage points versus the index and 9.68 percentage points versus the Bottom 25. The Top 25 also exceeded every one of the 100 sampled random portfolios. This combination makes 2019 the strongest evidence of model outperformance among the three cohorts.

The Bottom 25's substantial positive result remains part of that comparison. The model's strongest separation in this cohort was against the index and sampled random peers, while the high-score versus low-score spread was smaller. The result was strong in this cohort, but it does not show that the model will work equally well in other periods.



**Figure 2. Top 25 versus the mean of 100 random 25-stock portfolios: 36-month cumulative total returns.** Each random portfolio begins equally weighted and follows the same 36-month buy-and-hold rule. Source: Author calculations using Bloomberg Terminal data.

## 2022 cohort | January 2023–December 2025

The Top 25 returned 82.80% over 36 months, ahead of the Random Portfolio Mean of 65.28%, the S&P MidCap 400's 52.87%, and the Bottom 25's 46.00%. Recorded spreads were 29.93 percentage points versus the index and 36.80 percentage points versus the Bottom 25. The outperformance was meaningful economically, but the Top 25 exceeded only 73% of the sampled random portfolios.

The random-portfolio 10th–90th percentile range was 24.53%–105.64% for this cohort. The Top 25 result therefore occurred within a broad distribution of possible peer-portfolio outcomes. Its empirical p-value of 0.2772 does not indicate statistically unusual performance in that sampled distribution, despite its return above both the random mean and the index.

## 7 Statistical & Risk Analysis

### Empirical random-portfolio return test

The one-sided empirical test counts random portfolios with a 36-month cumulative total return at least as high as the Top 25. Its p-value is  $(1 + \text{that count}) / (100 + 1)$ . Ties count in the p-value, whereas the percentage beaten records only strict Top 25 outperformance. The adjustment makes  $1/101$ , approximately 0.0099, the smallest attainable p-value with 100 simulations.

| Cohort | Random portfolios beaten | Empirical p-value |
|--------|--------------------------|-------------------|
| 2014   | 16%                      | 0.8416            |
| 2019   | 100%                     | 0.0099            |
| 2022   | 73%                      | 0.2772            |

The 2019 outcome lies above all sampled random peers and provides the strongest statistical evidence in this study. The 2022 result is above most peers but is not statistically unusual in the sampled distribution. The 2014 result is weak relative to random selection. These p-values describe the cohort-specific sampled comparison; they do not measure the probability that the model will succeed in a future period or establish persistent alpha.

### Volatility and drawdown

Annualized volatility is the sample standard deviation of monthly portfolio returns ( $\text{ddof} = 1$ ), multiplied by the square root of 12. Maximum drawdown is the minimum ratio of wealth to its running peak, minus one, including initial wealth of one. Drawdowns use month-end observations and therefore do not capture intramonth extremes.

**Table 2. Annualized volatility and maximum drawdown**

| Cohort | Portfolio      | Annualized volatility | Maximum drawdown |
|--------|----------------|-----------------------|------------------|
| 2014   | Top 25         | 18.88%                | -28.94%          |
| 2014   | Bottom 25      | 17.36%                | -31.46%          |
| 2014   | S&P MidCap 400 | 11.15%                | -12.86%          |
| 2019   | Top 25         | 31.79%                | -32.95%          |
| 2019   | Bottom 25      | 32.68%                | -38.29%          |
| 2019   | S&P MidCap 400 | 25.89%                | -33.30%          |
| 2022   | Top 25         | 24.25%                | -18.89%          |
| 2022   | Bottom 25      | 28.00%                | -27.98%          |
| 2022   | S&P MidCap 400 | 15.85%                | -14.68%          |

Source: Author calculations using Bloomberg Terminal data. Volatility is annualized; drawdown is the largest observed month-end peak-to-trough loss.

The Top 25 was more volatile than the index in all three cohorts. In 2014, it combined weaker returns with a materially deeper drawdown than the index. In 2019, strong relative returns came with 31.79% volatility and a 32.95% drawdown, close to the index's 33.30% drawdown. In 2022, its 18.89% drawdown was deeper than the index's but smaller than the Bottom 25's. Return superiority in a cohort should therefore be assessed alongside the losses and variability experienced during the holding period.

## 8 Limitations

### Small sample and dependence on market environment

Three historical cohorts provide limited evidence about repeatability. The sharp contrast between 2014 and the later cohorts demonstrates that the model's observed performance varies across periods. These results cannot establish how reliably it would work across other formation dates or market environments. The empirical benchmark adds useful within-cohort context, but 100 random simulations do not substitute for a longer history of independent investment tests.

### Concentration and portfolio implementation

A 25-stock portfolio is concentrated. Individual holdings can materially influence cumulative returns, volatility, and drawdown, and buy-and-hold weight drift can increase that influence over time. Sector-neutral factor scores do not themselves ensure a sector-neutral investment portfolio. The reported outcomes therefore describe these particular portfolio construction rules, rather than a broadly diversified implementation of the signals.

The calculations exclude transaction costs, taxes, and other implementation frictions. The liquidity screen narrows the eligible universe, but it does not quantify execution costs or the capacity of a strategy at a particular investment size. The recorded returns consequently should not be read as net investor outcomes after real-world implementation.

### Historical inputs and company identities

Point-in-time historical screening is central to the design, but the analysis depends on the historical source data and reconstruction assumptions. Currency normalization, economic-company deduplication, historical GICS mappings, missing-factor treatment, and canonical-ticker return matching all affect the research dataset. Coverage and identity checks support those choices without eliminating every possible historical-data or corporate-identity limitation.

The study evaluates the full composite. It does not establish that revenue growth acceleration alone caused the observed results, quantify an independent acceleration effect, or demonstrate that each pillar contributed equally to successful selections. Likewise, cumulative return comparisons and the empirical peer-portfolio test do not establish persistent alpha. The findings remain conditional on the specified model, opportunity set, and three tested windows.

## 9 Conclusion

The 2019 cohort produced the strongest evidence of model outperformance. Its Top 25 earned 58.40% over 36 months, exceeded the index and Bottom 25, and outperformed all 100 sampled random peer portfolios. The empirical p-value of 0.0099 places it at the strongest attainable outcome under the 100-simulation test. One possible explanation is that the COVID-19 shock and subsequent recovery created unusually wide dispersion, rewarding companies with stronger growth and improving earnings expectations.

The 2022 cohort also delivered economically meaningful relative outperformance, with an 82.80% cumulative total return, although its empirical p-value of 0.2772 was not statistically unusual among the sampled random portfolios. That cohort began after a broad valuation reset caused by aggressive rate increases, which may have created a more favorable starting point for companies with improving fundamentals. By contrast, the 2014 cohort returned only 13.43% and lagged every primary comparison group, potentially reflecting a period in which falling energy and commodity prices and early monetary-policy tightening disrupted previously improving trends.

Overall, the evidence suggests that a composite emphasizing accelerating fundamentals, business quality, and growth-adjusted valuation may help identify mid-cap outperformers in certain market environments, but the results also appear sensitive to regime changes. Higher risk, concentrated portfolios, and the small historical sample limit the strength of that conclusion, and the available evidence is insufficient to establish a persistent investment edge.

## 10 Selected References

The model is informed by established research on momentum, fundamental signals, profitability, and multifactor stock selection. The following sources provide context for the investment characteristics evaluated in this study.

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Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1), 1–28. <https://doi.org/10.1016/j.jfineco.2013.01.003>

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## Appendix A. Exploratory Large-Cap Graduation Test

An earlier secondary analysis tested whether selected companies subsequently entered the Top 500 U.S. economic companies by market capitalization at least once during the 36-month evaluation window. This graduation outcome was exploratory and separate from the primary investment-return test.

Pooled Top 25 graduation was 14.67%, compared with 14.18% for the rest of the scoring universe. The pooled two-sided Fisher exact p-value was 0.8669. No meaningful pooled graduation effect was found. Realized investment returns are the more economically relevant primary outcome because crossing a market-cap rank threshold does not itself establish investment outperformance.

### Research and Data Sources

Analysis was performed in Python using the accompanying Jupyter notebook, with historical financial and market data sourced from Bloomberg Terminal. Licensed Bloomberg security-level data are not reproduced in this report.